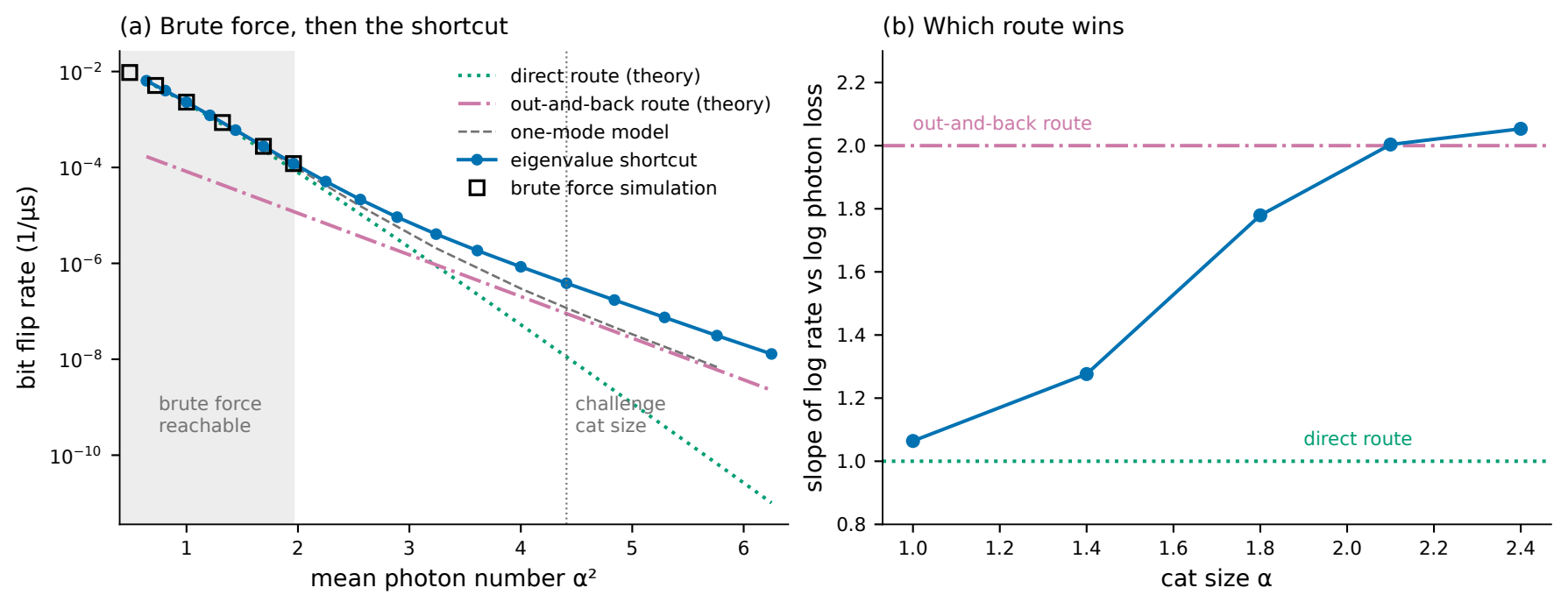
(a) Brute force, then the shortcut
direct route (theory)
out-and-back route (theory)
one-mode model
eigenvalue shortcut
brute force simulation
bit flip rate (1/μs)
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
brute force reachable
challenge cat size
1
2
3
4
5
6
mean photon number α²
(b) Which route wins
out-and-back route
direct route
slope of log rate vs log photon loss
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
cat size α